Machine Learning 2 (APM_50441_EP/DSAIB - 2026-2027). PC4

Exercise 1 (High dimensional curse)

Let $(X_i^{(j)})_{i,j\in\mathbb{N}}$ be a sequence of i.i.d. random variables with law $\mathcal{U}([0,1])$. For $i \in \mathbb{N}$ and $d \in \mathbb{N}^*$, denote X_i the random vector ${}^t\big(X_i^{(1)}, \dots, X_i^{(d)}\big)$. In the sequel, $\|\cdot\|$ denotes the usual Euclidean norm.

1. For $d \in \mathbb{N}^*$ and $i \in \mathbb{N}$, what is the law of X_i? For i, $i' \in \mathbb{N}$ and $i \neq i'$, prove that there exist $C \in \mathbb{R}$ and $C' \in \mathbb{R}$ to be specified such that
$$\frac{1}{d}\,\|X_i\|^2 \xrightarrow[d\to+\infty]{a.s.} C \qquad \text{and} \qquad \frac{1}{d}\,\|X_i - X_{i'}\|^2 \xrightarrow[d\to+\infty]{a.s.} C'.$$

2. Deduce that for a fixed $n \in \mathbb{N}^*$,
$$\frac{\max_{1\le i\le n}\|X_i\|^2}{\min_{1\le i\le n}\|X_i\|^2} \xrightarrow[d\to+\infty]{a.s.} 1.$$

3. In practice n and d often grow simultaneously. Obviously when n goes to $+\infty$ for a fixed d, the ratio will go to $+\infty$. We want to show that the ratio remains close to 1 as long as $\log(n) \ll d$, *i.e.* $\log(n) = o(d)$.
Prove that there exists $S > 0$ to be specified such that
$$\sqrt{d}\left(\frac{\|X_i\|^2}{d} - C\right) \xrightarrow[d\to\infty]{\mathcal{L}} \mathcal{N}(0\,,\,S).$$
Deduce that for $0 < \varepsilon < 1$, there exists $d_0 \in \mathbb{N}^*$ such that for $d \ge d_0$,
$$\mathbb{P}\Big(\big|\|X_i\|^2 - C\,d\big| > \sqrt{2\,d\,S\,\log(1/\varepsilon)}\Big) \le 4\,\varepsilon,$$
and thus
$$\mathbb{P}\Big(\sup_{1\le i\le n}\big|\|X_i\|^2 - C\,d\big| > \sqrt{2\,d\,S\,\log(n/\varepsilon)}\Big) \le 4\,\varepsilon$$
for $d \ge d_0'$, where d_0' may depend on n.

4. Explain why this is sufficient to justify heuristically that ratio between the maximum and the minimum tends to 1 as long as $\log(n)/d \to 0$.

5. Using the Hoeffding Inequality (2), prove that the ratio converges to 1 in probability as soon as $\log(n)/d \to 0$.

6. *Pairwise distances.* Prove that the ratio between the largest and the smallest of the $n(n-1)/2$ distances $\|X_i - X_{i'}\|$, $i \neq i'$, also converges to 1 in probability as soon as $\log(n)/d \to 0$. Why does this make the nearest neighbors meaningless in high dimension? (This is the theorem of the lecture, and Lab 4 checks numerically that $\|X_i\|/\sqrt{d} \to \sqrt{1/3}$ and $\|X_i - X_{i'}\|/\sqrt{d} \to \sqrt{1/6}$.)

Hints. You may use the following concentration inequality for $U \sim \mathcal{N}(0,1)$ a standard Gaussian distribution
$$\mathbb{P}(|U| > t) \le 2\,\exp\big(-\frac{1}{2}\,t^2\big), \tag{1}$$
and the Hoeffding's inequality which states that, given $(Z_i)_{1\leqslant i\leqslant n}$, n independent random variables such that for all $1 \leqslant i \leqslant n$, $\mathbb{P}(a_i \leqslant Z_i \leqslant b_i) = 1$ where a_i, b_i are real numbers such that $a_i < b_i$, for all $t > 0$,
$$\mathbb{P}\Big(\Big|\sum_{i=1}^n \big(Z_i - \mathbb{E}\,[Z_i]\,\big)\Big| > t\Big) \leqslant 2\exp\Big(\frac{-2\,t^2}{\sum_{i=1}^n (b_i - a_i)^2}\Big)\,. \tag{2}$$

Exercise 2 (Kernel Principal Component Analysis)

Let $(X_i)_{1\le i\le n}$ be n observations in a general space $\mathcal{X}$, $k : \mathcal{X}\times\mathcal{X} \to \mathbb{R}$ a positive definite kernel and $\mathbf{K} = (k(X_i, X_j))_{1\leqslant i,j\leqslant n}$. $\mathcal{W}$ denotes the Reproducing Kernel Hilbert Space associated with k and for all $x \in \mathcal{X}$, $\phi(x)$ denotes the function $\phi(x) : y \mapsto k(x,y)$. The aim is now to perform a PCA on $(\phi(X_1), \dots, \phi(X_n))$. It is assumed that $\sum_{i=1}^n \phi(X_i) = 0$.

1. Prove that
$$f_1 = \operatorname*{argmax}_{f\in\mathcal{W}\,;\,\|f\|_{\mathcal{W}}=1} \sum_{i=1}^{n} \langle \phi(X_i), f\rangle_{\mathcal{W}}^2$$
may be written
$$f_1 = \sum_{i=1}^{n} \alpha_1(i)\phi(X_i)\ , \quad \text{where} \quad \alpha_1 = \operatorname*{argmax}_{\alpha\in\mathbb{R}^n\,;\,\alpha^\top \mathbf{K}\alpha=1} \alpha^\top \mathbf{K}^2\alpha\ .$$

2. Prove that $\alpha_1 = \lambda_1^{-1/2} b_1$ where b_1 is the unit eigenvector associated with the largest eigenvalue λ_1 of $\mathbf{K}$. What about $(f_2, \ldots, f_r)$, with $r = \operatorname{rank}(\mathbf{K}) \leq n$, defined iteratively by
$$f_1 \in \operatorname*{argmax}_{f\in\mathcal{W}\,;\,\|f\|_{\mathcal{W}}=1} \sum_{i=1}^{n} \langle \phi(X_i), f\rangle_{\mathcal{W}}^2 \quad \text{and for all } \ 2\leqslant k\leqslant r\ , \quad f_k \in \operatorname*{argmax}_{\substack{f\in\mathcal{W}\,;\,\|f\|_{\mathcal{W}}=1\,;\\ f\perp f_1,\ldots,f\perp f_{k-1}}} \sum_{i=1}^{n} \langle \phi(X_i), f\rangle_{\mathcal{W}}^2 ?$$

3. Write $H_q = \operatorname{span}\{f_1, \ldots, f_q\}$. Prove that, for all $1\leqslant i\leqslant n$,
$$\pi_{H_q}(\phi(X_i)) = \sum_{j=1}^{q} \lambda_j \alpha_j(i) f_j\ .$$

4. *Centering.* In general $\sum_i \phi(X_i) \neq 0$. Show that the kernel matrix of the centered features $\tilde{\phi}(X_i) = \phi(X_i) - \frac{1}{n}\sum_l \phi(X_l)$ is $\tilde{\mathbf{K}} = \mathbf{HKH}$ with $\mathbf{H} = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, so that the previous results apply to $\tilde{\mathbf{K}}$.

5. *New points and MDS.* Give the coordinate $\langle \phi(x), f_j\rangle_{\mathcal{W}}$ of a new point x on the j-th component (with the centered features, assuming $\sum_i \phi(X_i) = 0$). Show that the coordinates of the observations, $\left(\lambda_j^{1/2} b_j(i)\right)_{j\leqslant q}$, are the ones given by classical multidimensional scaling (MDS) applied to the matrix of squared distances $\|\phi(X_i) - \phi(X_j)\|_{\mathcal{W}}^2$. How is ISOMAP related to kernel PCA?

Exercise 3 (Principal component analysis: three criteria, one solution)

Let $\mathbb{X}$ be the $n \times d$ matrix of centered observations $x_1, \ldots, x_n \in \mathbb{R}^d$ (rows), and $\mathbb{X} = UDV^\top$ its singular value decomposition ($d_1 \geqslant d_2 \geqslant \cdots \geqslant 0$). For $q \leqslant d$, we look for a linear subspace E of dimension q.

1. *Inertia and reconstruction.* With π_E the orthogonal projection on E, show that maximizing the inertia of the projected points $\sum_i \|\pi_E(x_i)\|^2$ is equivalent to minimizing the reconstruction error $\sum_i \|x_i - \pi_E(x_i)\|^2$.

2. For $q = 1$ and $E = \mathbb{R}v$ with $\|v\| = 1$, show that the inertia is $v^\top \mathbb{X}^\top \mathbb{X} v$, and that it is maximal for the first right singular vector v_1 (the first eigenvector of $\mathbb{X}^\top\mathbb{X}$), with the value d_1^2. What are the coordinates (principal components) of the observations?

3. *Relationship criterion.* Show that the same subspace preserves the scalar products between observations as well as possible: $U_q D_q$ minimizes $\|\mathbb{X}\mathbb{X}^\top - ZZ^\top\|_F$ among the $n \times q$ matrices Z (you may admit the Eckart–Young theorem: the best rank-q approximation of a matrix in Frobenius norm is its truncated SVD). Deduce that the reconstruction error of the best subspace is $\sum_{j>q} d_j^2$, and relate it to the "explained variance" plots of Lab 4.

Exercise 4 (Gaussian mixtures: EM and the ELBO)

A Gaussian mixture has a latent variable $Z \in \{1, \ldots, K\}$ with $\mathbb{P}(Z = k) = \pi_k$ and $X|Z = k \sim \mathcal{N}(\mu_k, \Sigma_k)$; we write $\theta = (\pi_k, \mu_k, \Sigma_k)_k$ and φ_k the Gaussian densities.

1. *ELBO.* For any law q on $\{1, \ldots, K\}$, prove that
$$\log p_\theta(x) = \underbrace{\sum_k q(k) \log \frac{p_\theta(x,k)}{q(k)}}_{\mathsf{ELBO}(q,\theta)} + \mathrm{KL}\left(q, p_\theta(\cdot|x)\right)\ ,$$
where $p_\theta(x,k) = \pi_k\varphi_k(x)$ and $p_\theta(k|x) = p_\theta(x,k)/p_\theta(x)$. Deduce that the ELBO is a lower bound of the log-likelihood, with equality for $q = p_\theta(\cdot|x)$.

2. *EM.* Given observations $x_1, \dots, x_n$, the EM algorithm alternates: (E) $q_i = p_{\theta^{(t)}}(\cdot|x_i)$, the *responsibilities* $r_{ik} = \pi_k \varphi_k(x_i) / \sum_l \pi_l \varphi_l(x_i)$; (M) $\theta^{(t+1)}$ maximizes $\sum_i \mathrm{ELBO}(q_i, \theta)$. Show that the log-likelihood never decreases along the iterations, and compute the M step for π_k and μ_k.

3. What do the E and M steps become when $\Sigma_k = \sigma^2 I$ with $\sigma \to 0$ and $\pi_k = 1/K$? Relate this to k-means. Why do the VAE of the lecture also maximize an ELBO?

Exercise 5 (k-means and Ward's agglomerative clustering)

For a partition $C_1, \dots, C_K$ of $x_1, \dots, x_n \in \mathbb{R}^d$ with centers $\mu_1, \dots, \mu_K$, the *within-cluster inertia* is $W = \sum_k \sum_{i \in C_k} \|x_i - \mu_k\|^2$.

1. Lloyd's algorithm alternates the assignment of each point to its closest center and the update of each center to the mean of its cluster. Prove that W does not increase at each step, and deduce that the algorithm stops after a finite number of iterations. Does it find the global minimum?

2. Ward's method merges, at each step, the two clusters whose fusion increases W the least. Show that merging A and B (of sizes n_A, n_B and means μ_A, μ_B) increases W by $\frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2$.

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